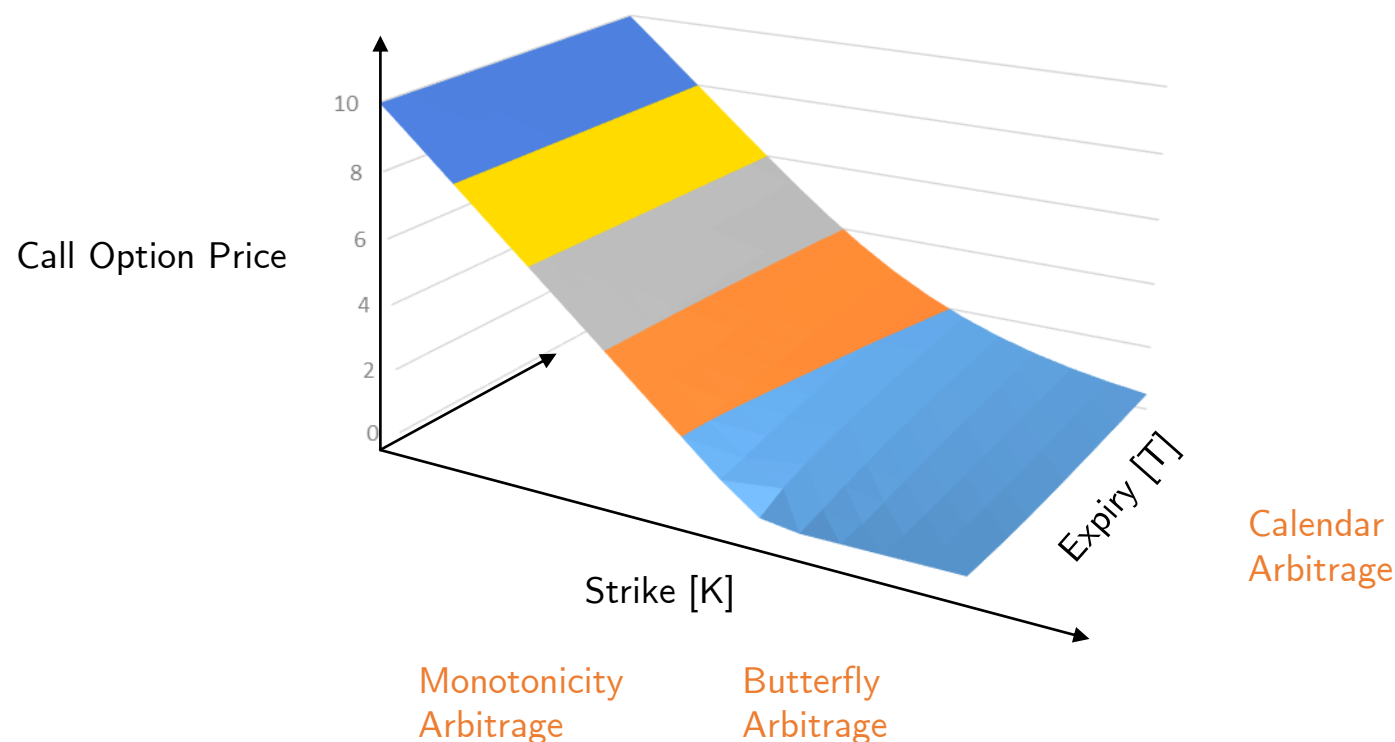


Volatility Models

(3. Smile Calibration - SABR)

Arbitrage-Free conditions

Market quotes (prices or IV) are available for only a discrete set of options, and we must rely on interpolation/extrapolation techniques to build a continuum of prices. But we need to do it ensuring arbitrage-free interpolation. Arbitrage can occur in the time as well as in the strike direction.



Arbitrage-Free conditions



Mathematically speaking,

$$\frac{\partial V_C}{\partial T} > 0 \quad V_C(t_0, S_0; K, T + \Delta T) - V_C(t_0, S_0; K, T) > 0 \quad \text{calendar spread condition}$$

$$\frac{\partial V_C}{\partial K} < 0 \quad V_C(t_0, S_0; K + \Delta K, T) - V_C(t_0, S_0; K, T) < 0 \quad \text{monotonicity condition}$$

$$\frac{\partial^2 V_C}{\partial K^2} > 0 \quad V_C(t_0, S_0; K + \Delta K, T) - 2V_C(t_0, S_0; K, T) + V_C(t_0, S_0; K - \Delta K, T) < 0 \quad \text{butterfly condition}$$

Fitting the smile

So, we have range of market quotes available for various strikes K_j and for each maturity T_i . There are different techniques available like linear, spline and other parametric methods. **Linear interpolation** in the total variance space $w(K, T) = \sigma_{imp}^2(K, T)T$ on maturity dimension is common practice. But blind linear interpolation in both strike and maturity dimension is likely to violate no-arbitrage conditions.

One of the popular parametric interpolation of smile is given by Hagan *et al.* [2002]. It has a direct linkage to SABR model (a popular stochastic volatility model). This parametrization is popular because Black-Scholes implied volatilities are analytically available (below)

$$\hat{\sigma}(T, K) = \frac{\hat{a}(K)\hat{c}(K)}{g(\hat{c}(K))} \times \left[1 + \left(\frac{(1-\beta)^2}{24} \frac{\alpha^2}{(S_F(t_0)K)^{1-\beta}} + \frac{1}{4} \frac{\rho\beta\gamma\alpha}{(S_F(t_0)K)^{\frac{1-\beta}{2}}} + \frac{2-3\rho^2}{24} \gamma^2 \right) T \right]$$

$$\text{where } \hat{a}(K) = \frac{\alpha}{(S_F(t_0)K)^{\frac{1-\beta}{2}} \left(1 + \frac{(1-\beta)^2}{24} \log^2 \left(\frac{S_F(t_0)}{K} \right) + \frac{(1-\beta)^4}{1920} \log^4 \left(\frac{S_F(t_0)}{K} \right) \right)}$$

$$\hat{c}(K) = \frac{\gamma}{\alpha} (S_F(t_0)K)^{\frac{1-\beta}{2}} \log \frac{S_F(t_0)}{K} \quad g(x) = \log \left(\frac{\sqrt{1-2\rho x + x^2} + x - \rho}{1-\rho} \right)$$

Impact of parameters

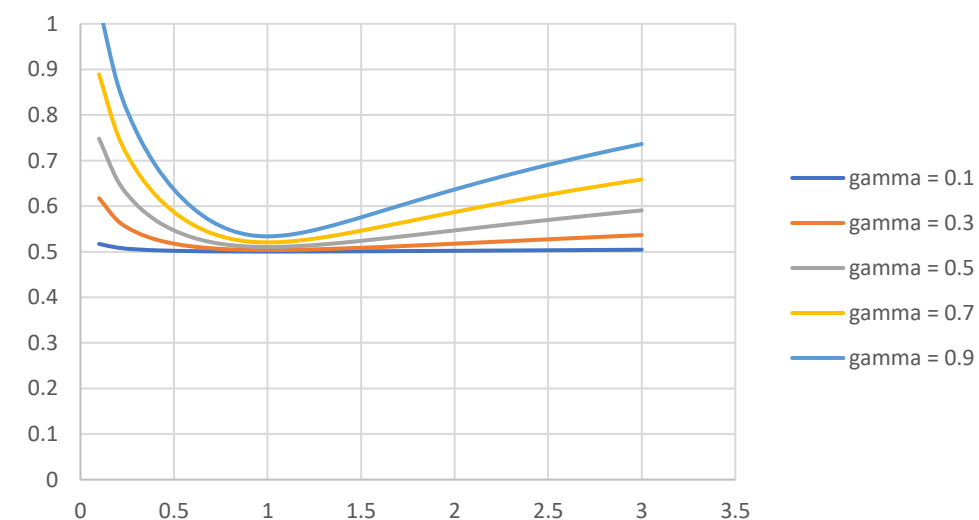
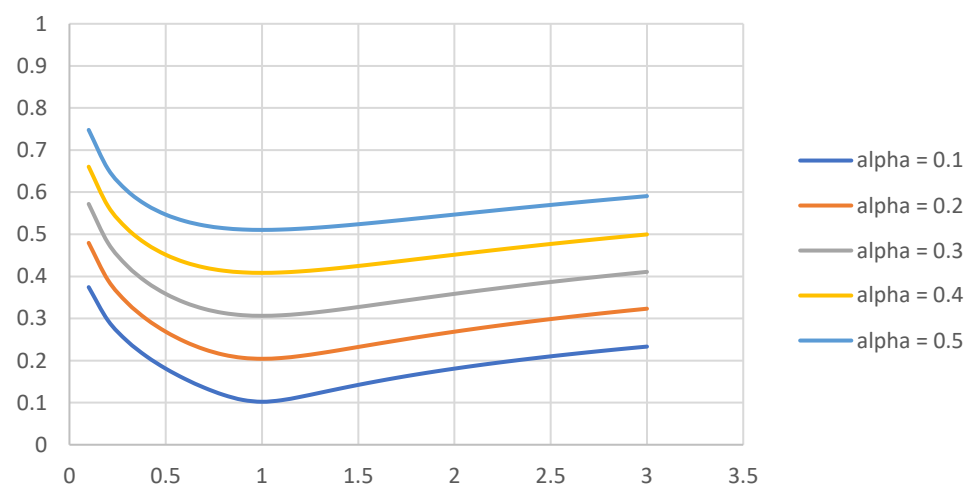
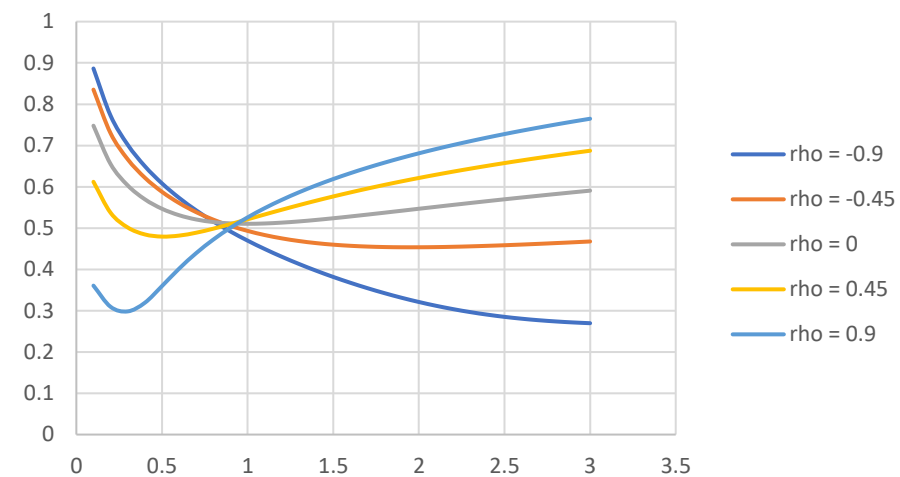
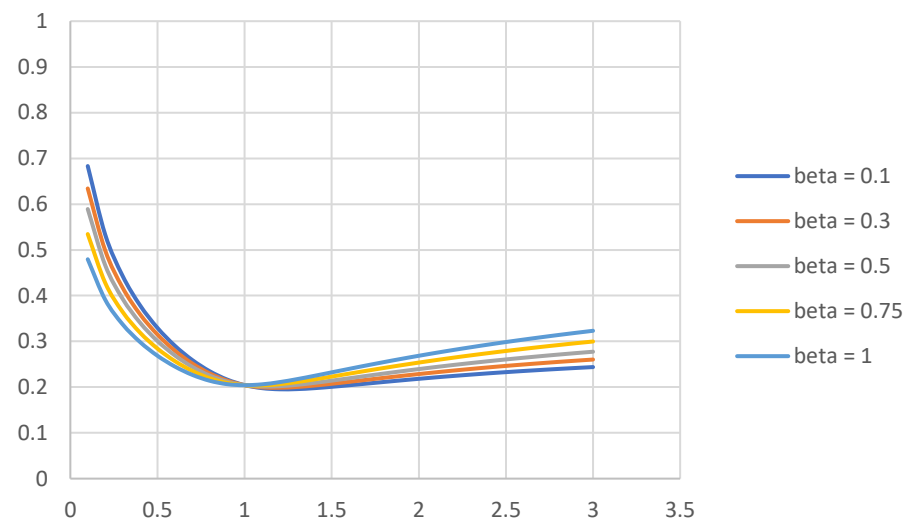
$$dF_t = \alpha_t F_t^\beta dW_t^{(1)} \quad d\alpha_t = \gamma \alpha_t dW_t^{(2)} \quad dW_t^{(1)} dW_t^{(2)} = \rho dt \quad \text{SABR Model}$$

Both parameters β and ρ impact the implied volatility skew. In practice, often β is fixed, whereas ρ is used in calibration. Parameter α controls the level of implied volatility smile and γ controls the curvature of the smile. For ATM options, $S_F(t_0) = K$

$$\hat{\sigma}(T, K) \approx \frac{\alpha}{(S_F(t_0)K)^{1-\beta}} \left[1 + \left(\frac{(1-\beta)^2}{24} \frac{\alpha^2}{(S_F(t_0))^{2-2\beta}} + \frac{1}{4} \frac{\rho\beta\gamma\alpha}{(S_F(t_0)K)^{1-\beta}} + \frac{2-3\rho^2}{24} \gamma^2 \right) T \right] \quad \text{Derive it!}$$

Let's visualize the smile for various parameters

Impact of parameters



Smile Interpolation - Example

Consider these IV market quotes for IBM for $T = 1$ and $F = 131.5$

T=1y	K=110.0	K=121.3	K=131.5	K=140.9	K=151.4
$\sigma_{imp}(T, K)$	14.2	11.8	10.3	10.0	10.7

Let's experiment with 2 interpolation schemes

- Linear
- Hagan

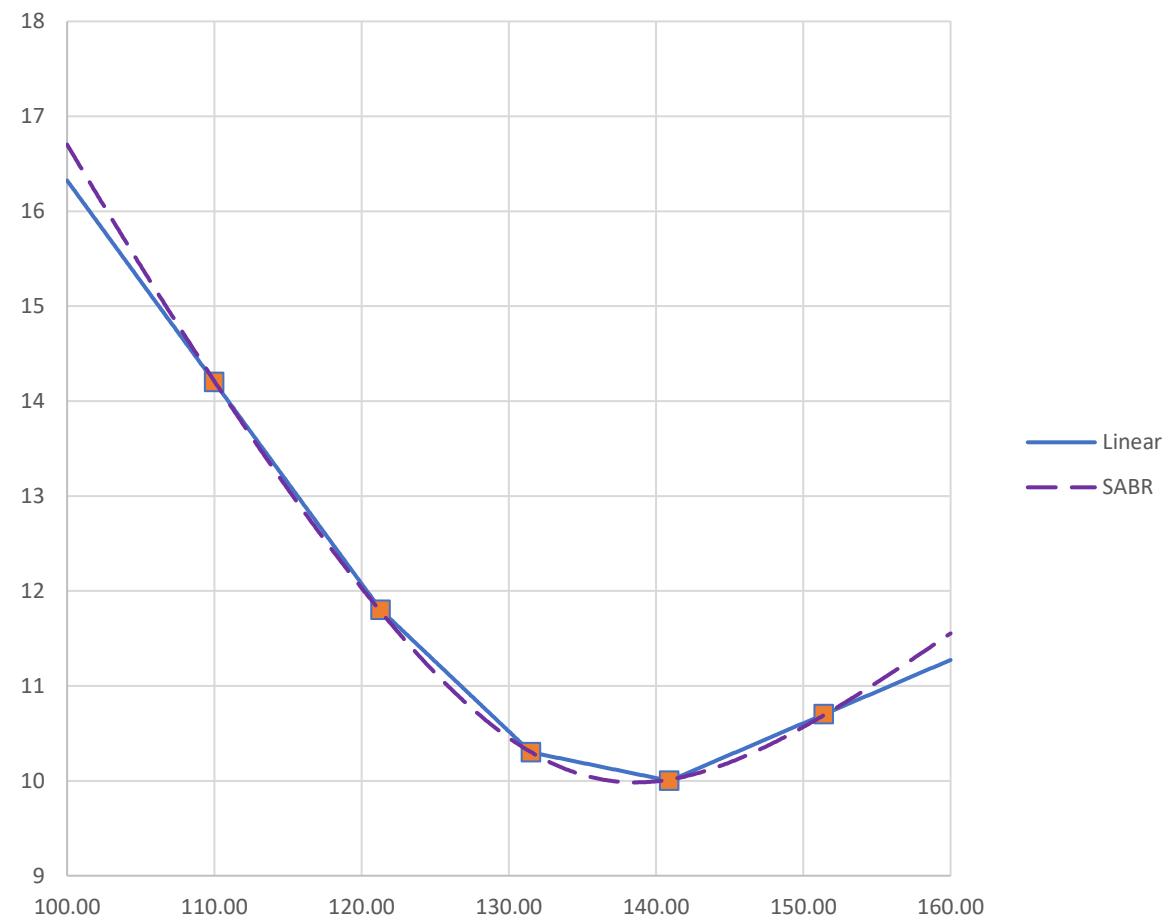
To calibrate Hagan's parameters, we minimize the sum of squared differences between observed and implied vols

$$\min_{\alpha, \beta, \gamma, \rho} \sum_{K_i} \left(\hat{\sigma}(T, K_i) - \sigma_{imp}(T, K_i) \right)^2$$

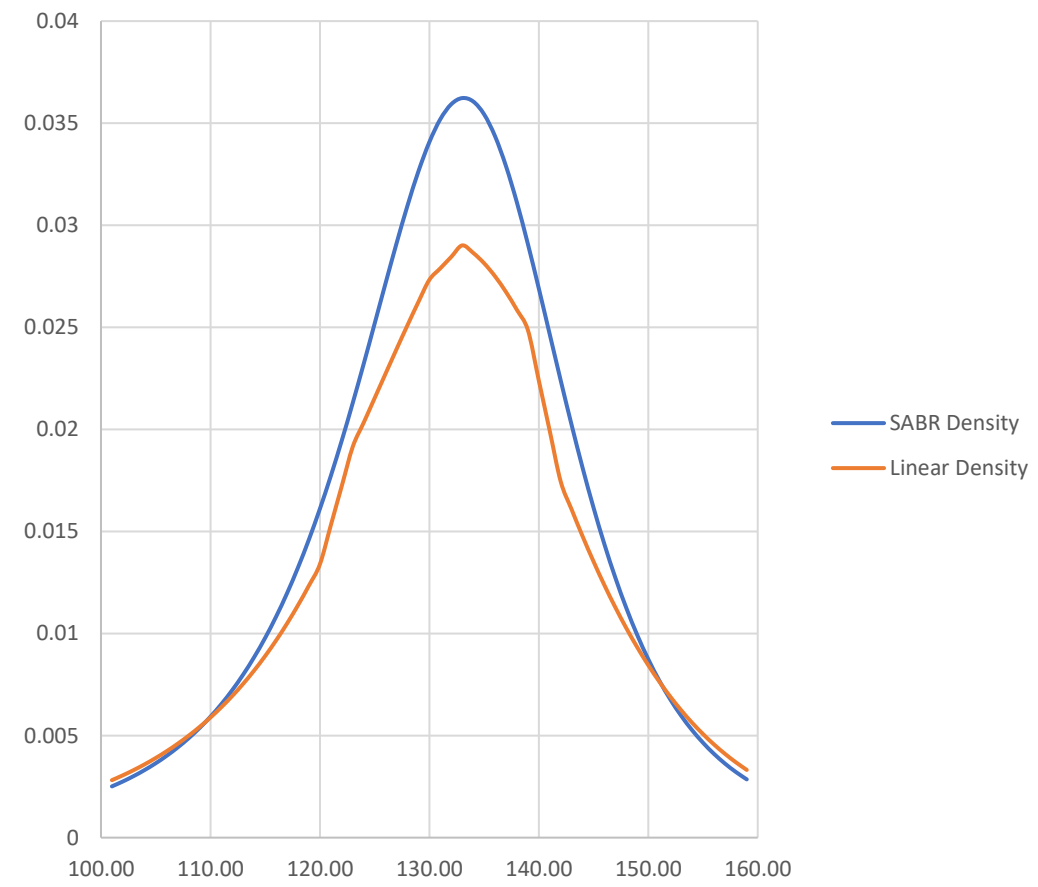
Smile Interpolation - Example



Interpolation



Risk Neutral Density



Fitting the term structure



Typically, in practice, we fix β for the whole asset globally. We can then calibrate (α, γ, ρ) for each maturity slice (T). Then we can use linear interpolation in the maturity dimension (total implied variance)